Ex.No.11 Date: 03.09.2026

Routing a network layer.

a) Simulate static routing protocal configuration

b) Simulate RIP

Aim:

To simulate static routing protocal configuration and RIP using cisco packet tracer.

Algorithm / Procedure:

Part - A (Static Routing)

* open cisco packet tracer and create the given network topology with routers.

* configure the networks available on the router interfaces

* open the router CII and enter the required configuration mode.

* configure the first route as the main route and the second route as the backup route

* use the tracert command to trace the path taken by the packets.

* To test the backup route, remove the link between the routers and observe the change in the routing path.

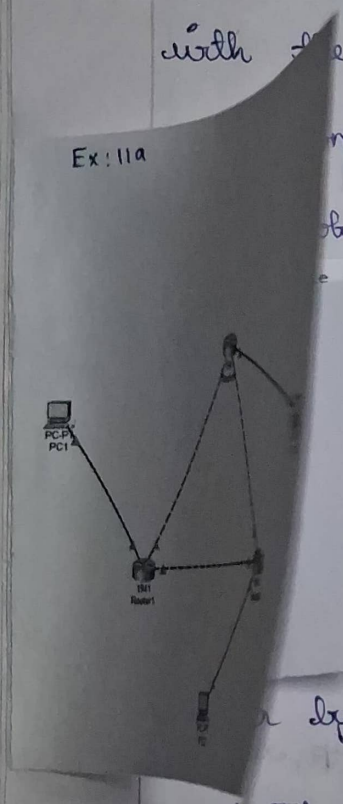
* configure the backup static route and verify that the packet is transmitted through the alternate route.

Part B - RIP using cisco packet tracer.

* configure the IP address of the PCs and routers interfaces according to the given network

* configure the serial interfaces of the routers with the given IP address.

Ex: 11a



nfigure and enable the RIP routing protocol

bal configuration mode.

onnected network that hav

d to test connectivity

and to observe the route

by the packets.

* If the [illegible] route fails, RIP automatica

le route for packet transmis

Ex: 11b

```
Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=55ms TTL=125
Reply from 20.0.0.2: bytes=32 time=56ms TTL=125
Reply from 20.0.0.2: bytes=32 time=60ms TTL=125
Reply from 20.0.0.2: bytes=32 time=55ms TTL=125

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 55ms, Maximum = 60ms, Average = 56ms

C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      20.0.0.1
  2   11 ms      8 ms      5 ms      192.168.1.250
  3   29 ms     26 ms     25 ms      192.168.1.245
  4   22 ms     28 ms     20 ms      20.0.0.2

Trace complete.
```

and RIP were successfully

packet tracer and packet

routing was verified